COURSEWORK ASSIGNMENT SUBMISSION

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Subject: Software Engineering & Project Management **Day:** Day 14 — Requirements Validation, Feasibility & Management

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Section 12: Classroom Activity — Validate and Manage a Change

Scenario: Following baseline SRS sign-off, campus coordinators submit an urgent Change Request: "Allow registered students to cancel their bookings and automatically re-allocate seats to waitlisted students."

1. Evaluation of Raw Request

- **Deficiencies in Raw Statement:** Does not specify the cancellation cutoff deadline, cancellation quota restrictions, notification triggers, or how long waitlisted students have to claim reopened seats.
- **Clarification Inquiries:** (1) Up to what time prior to event start is cancellation permitted? (2) Does the system automatically assign the seat, or send an invitation link with a 2-hour claim window? (3) Can a student re-register after cancelling?

2. Multidimensional Feasibility Study

Feasibility Dimension	Feasibility Rating	Technical / Operational Findings
Technical Feasibility	Feasible (Low Risk)	Existing PostgreSQL database supports transactional status transitions; background queue can easily handle waitlist promotion jobs.
Economic Feasibility	Feasible (Zero Cost)	Requires no additional commercial software licenses or cloud infrastructure expansion; fits within current developer capacity.
Operational Feasibility	Highly Feasible	Significantly reduces empty venue seats and eliminates manual coordinator intervention to fill cancelled slots.
Schedule Feasibility	Moderate Risk	Requires 4 days of engineering effort (backend state machine, email worker, UI update, and regression testing); fits into Sprint 2 buffer.
Legal / Policy Feasibility	Fully Compliant	Complies with university event policy granting equal registration access without monetary liability.

3. Change Control Board (CCB) Recommendation

Recommendation: APPROVE with scope containment. The feature shall be implemented for single-ticket events with a strict 24-hour pre-event cancellation cutoff. Seats will be automatically allocated to the next waitlisted student, who is granted a 4-hour window to confirm before the seat rolls to the next candidate.

Section 15: Homework — Validation & Change-Control Packet

Formal Change-Management protocol and impact analysis for CEMS.

1. Formal Change Request Specification (CR-2026-07)

- **Change Request Title:** Integration of Dynamic Seat Cancellation & Automated Waitlist Cascade.
- **Originator:** Vice President, Student Council & Festival Head.
- **Detailed Technical Impact:** (a) Database: Add `WAITLISTED`, `OFFERED`, and `CANCELLED` states to `registration_status` enum; (b) Backend: Introduce scheduled cron job `WaitlistCascadeWorker` to poll expired offers; (c) Notifications: Add high-priority email dispatch template; (d) Frontend: Add "Cancel Booking" action with modal confirmation on student dashboard.

2. Standard Change-Control Procedure (5 Stages)

Stage	Action Item	Responsible Party	Mandatory Output
1. Request Logging	Submit formal Change Request Form detailing problem and proposed change.	Requesting Stakeholder	Logged CR Entry in Jira
2. Impact Analysis	Evaluate architectural, schedule, cost, test, and security impact.	Lead Software Architect	Technical Impact Report
3. Formal Review	Convene CCB to approve, reject, defer, or request modification.	Change Control Board	Signed CCB Decision Log
4. Implementation	Develop code branch, update design documents, and write unit/regression tests.	Assigned Developers	Merged Git PR & Test Logs
5. Baseline Update	Update SRS version numbers, RTM mapping, and notify affected teams.	Requirements Analyst	Updated SRS v1.2 & Release Note